ScholarReady

Scholarship document readiness

Digital Image Processing and Data Science
Project 042

Sakshi Gupta and Jigisha Baiswar
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Local evidence and readiness

1 Image or PDF ingestion and original quality measures
2 Selected DIP branch and local Tesseract OCR
3 Document class and spatial field evidence
4 JSON requirements and unresolved issues
5 Shared CLI, evaluation and Streamlit result

Printed English, five supported types and other
Fictional checklist with explicit category applicability

Synthetic data and grouped partitions

Partition	Groups	Sources	Pages
Train	10	60	130
Validation	4	24	52
Test	4	24	52

All test references checked before inference
24 correlated bundle scenarios across four applicants
One reserved layout family is a close template variant

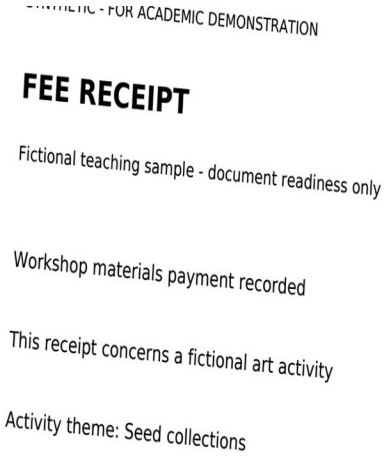
Fixed image processing branches

Branch	Enabled operations
A	Ingested original
B	Grayscale and Otsu threshold
C	Grayscale, deskew, CLAHE, median, adaptive threshold, opening then closing

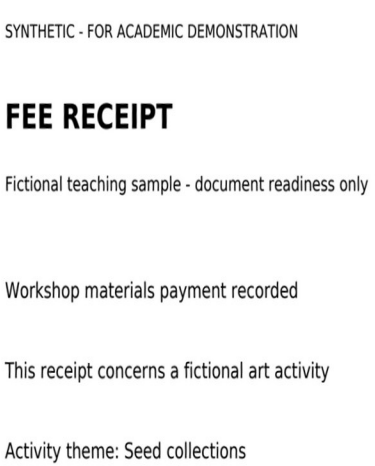
Same 52 pages and OCR settings
Tesseract 5.3.4, eng, OEM 3, PSM 3
300 DPI PDF rendering and 20 second OCR timeout

Saved processing stages

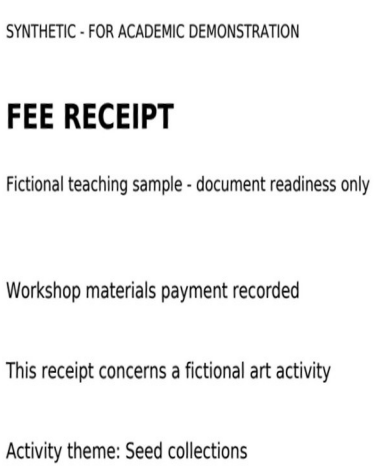
Original



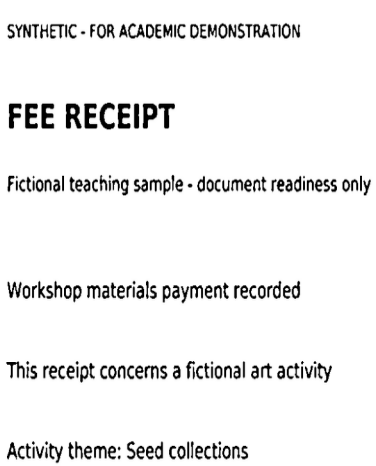
Deskew



CLAHE



Adaptive



Four of seven saved stages, receipt exp-16-5-rotation
A has 0 character edits, C has 4
Darker text does not establish better OCR

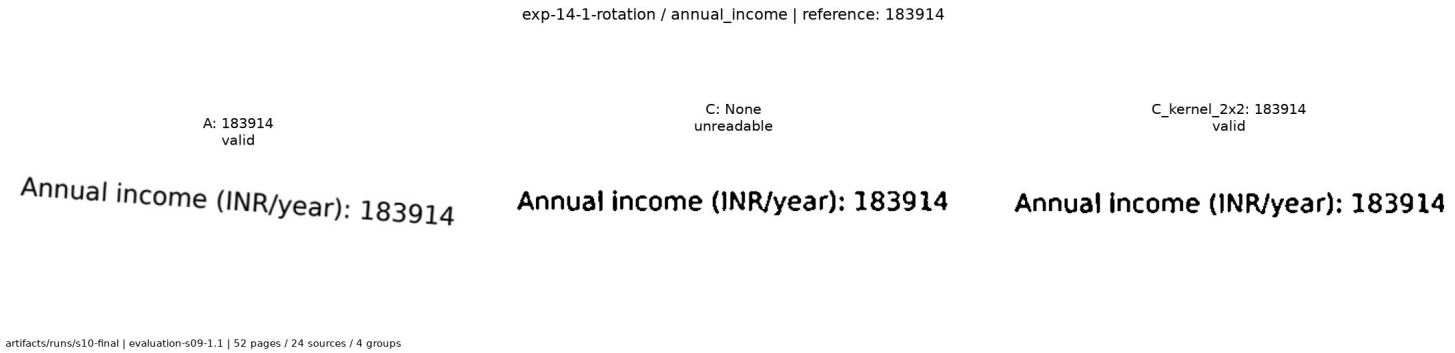
C increased OCR errors on this test

Route	Char edits	Word edits	Fields /164
A/keyword	3	4	161
A/learned	3	4	161
B/keyword	22	22	153
C/keyword	50	46	158

14,900 reference characters and 2,092 words
A/B/C CER 0.0201% / 0.1477% / 0.3356%
52 synthetic pages from 24 sources and four groups

Classification and field evidence

Keyword macro-F1 0.944444
Learned A macro-F1 1.000000
Other recall rises from 4/8 to 8/8



A extracts 183914 INR/year
C leaves the value null after its token-score check
Model and OCR scores remain uncalibrated

Every predeclared ablation remains visible

Variant	Edits	Fields
A	3	161
B	22	153
C	50	158
No deskew	37	159
No CLAHE	165	160
No filter	35	161

Variant	Edits	Fields
No threshold or morphology	0	164
No morphology	50	158
Grayscale only	3	161
Kernel 1 by 2	83	158
Kernel 2 by 1	111	158
Kernel 2 by 2	158	142

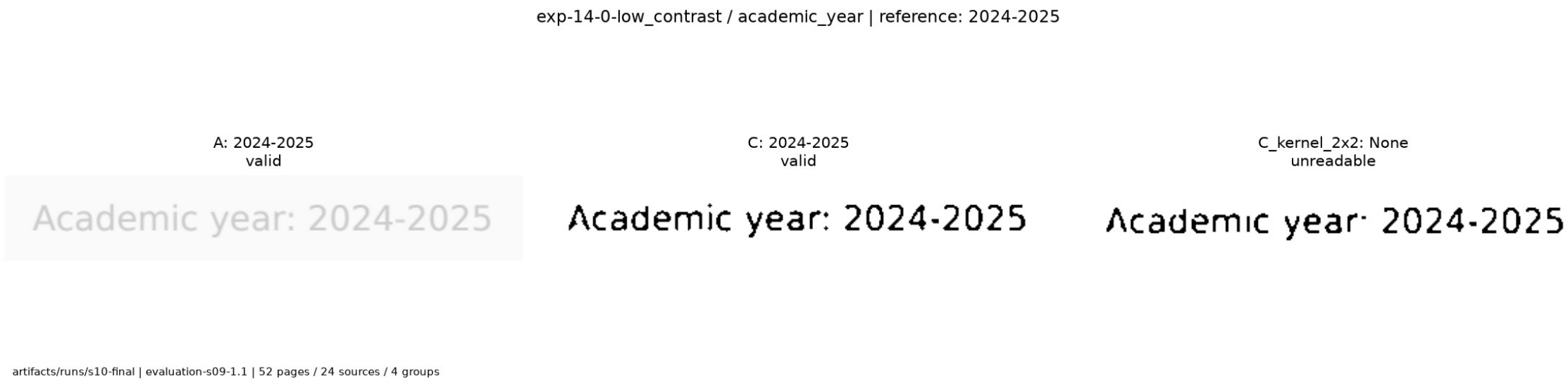
Character edits / 14,900 and exact fields / 164
Zero-edit diagnostic needs a new untouched evaluation

False ready is zero in 16 nonready bundles

Route	Headlines /24	Predicted ready	False ready
A/keyword	24	8	0/16
A/learned	24	8	0/16
B/keyword	22	6	0/16
C/keyword	24	8	0/16

Missing precision, recall and F1 are 1.0 on 120 items
Kernel 2 by 2 still agrees on only 18/24 headlines
Correlated synthetic cases do not establish real-world risk

Morphology can discard readable fields



Kernel 2 by 2 rejects the readable academic year
Stroke changes can also corrupt numeric values
Raw text, boxes and uncertainty stay inspectable

Four minute fictional demonstration

Case	Automatic headline	Completeness
Complete	ready	100%
Missing income	incomplete	80%
Unreadable amount	needs_clearer_upload	80%
Applicant mismatch	needs_review	0%

Run scripts/run_demo.py with a new output folder
Inspect result.json, field evidence and item states
No manual correction or hidden reference input

Reproduction and remaining limits

Smoke runs real development OCR
Final mode verifies preserved OCR and all 13 routes
Final test consumed remains YES

Synthetic English pages and closely related templates
Four final applicant groups and correlated bundles
Heuristic uncertainty and single-agent reference review
Student review and administrative details remain pending